Spatial Genomics

User Guide:

Sample Processing for RNA Profiling

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Sample Preparation Overview

This user guide is intended for downstream processing of fresh frozen, fixed frozen, and FFPE (Formalin Fixed Paraffin Embedded) samples after tissue collection. Follow the tissue collection guide relevant to your tissue type before starting these steps.

A. RNA Stabilization

After initial processing appropriate to your sample's tissue type, the sample is treated with stabilization buffers included in the kit to stabilize the RNA.

B. Pre-Hydrogel Embedding

The tissue sample is treated prior to hydrogel embedding.

C. Hydrogel Embedding

The tissue sample is embedded in a hydrogel polymer solution to protect the tissue structure.

D. Tissue Digestion

The hydrogel-tissue hybrid is subjected to a tissue digestion step for removal of light scattering molecules throughout the sample.

E. Primary Probe Hybridization

Primary probes are hybridized to the digested sample in a hybridization mix.

F. Post-Hybridization Wash

Excess primary probes are washed away with high stringency wash buffer to achieve optimal signal-to-noise fluorescence signal.

G. Probe Securing

The primary probes are secured with probe securing solutions before amplification.

H. Amplification

The sample is processed on the SG Booster. At the end of this step the sample is ready to be set up for imaging on the GenePS.

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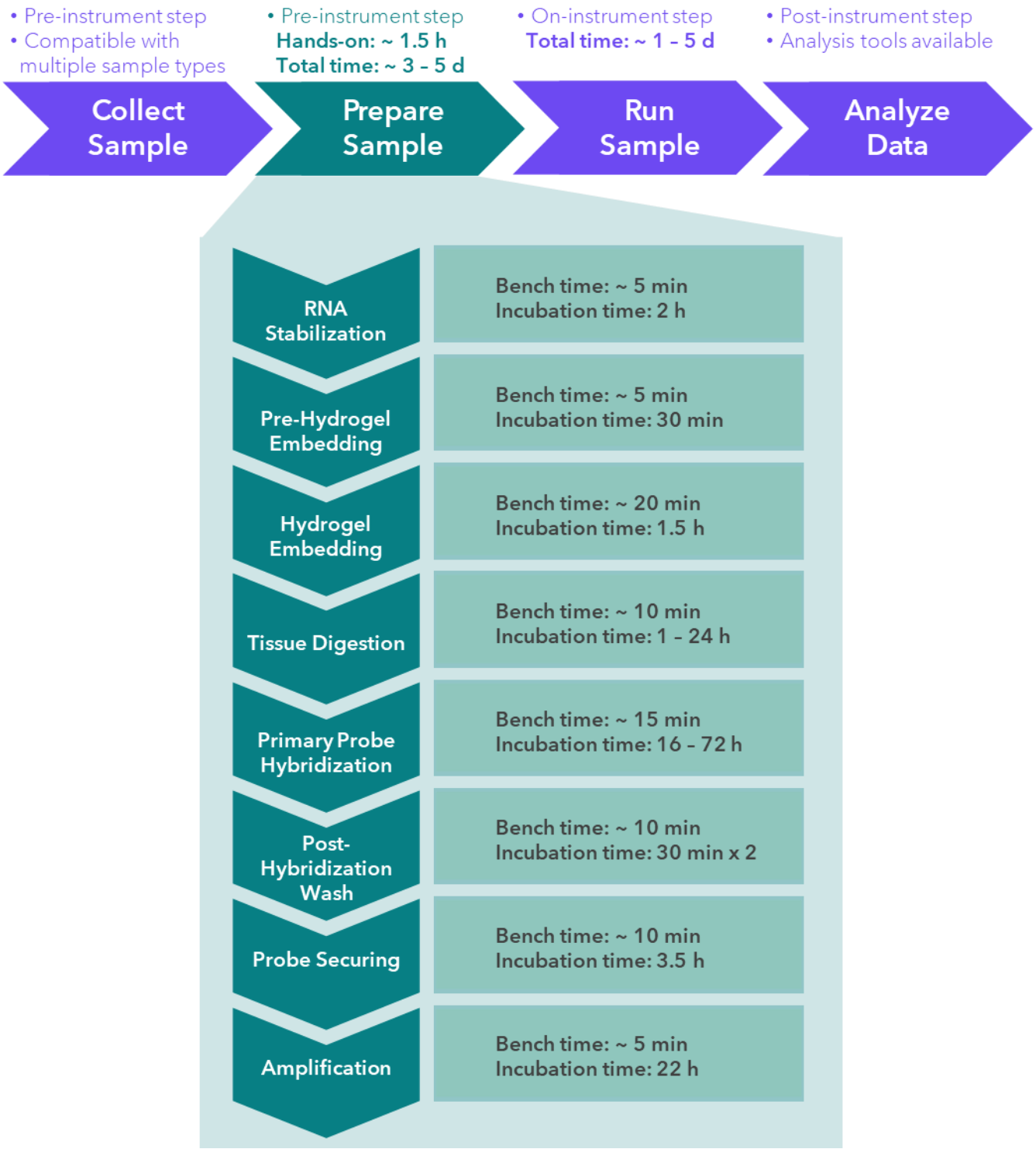
Schematic Overview



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Example Timetables and Kits

General Timetable

	Protocol steps	Hands-on time	Waiting time	Expression profiling kits	Validation check kits
Day 0	• Tissue Sectioning	(variable)			
Day 1	• RNA Stabilization • Pre-Hydrogel Embedding • Hydrogel Embedding • Tissue Digestion	5 min 5 min 20 min 10 min	2 h 30 min 1.5 h 1 – 24 h	Sample Preparation Kit	Sample Preparation Kit
Day 2	• Primary Probe Hybridization	15 min	16 – 72 h	Primary Probes Kit	Validation Kit
Day 3	• Post-Hybridization Wash • Probe Securing • Amplification	10 min 10 min 5 min	1 h 3.5 h 22 h	Amplification Kit	
Day 4	• Imaging	30 min	(variable)	GenePS Kit	

Example Timetable: 4-day protocol

Day 0		Tissue collection
Day 1	Monday	RNA stabilization Pre-hydrogel embedding Hydrogel embedding Tissue digestion – 16 h
Day 2	Tuesday	Primary probe hybridization – 24 h
Day 3	Wednesday	Post-hybridization wash Probe securing Amplification
Day 4	Thursday	Image on GenePS

Example Timetable: 6-day protocol

Day 0		Tissue collection
Day 1	Thursday	RNA stabilization Pre-hydrogel embedding Hydrogel embedding Tissue digestion – 16 h
Day 2	Friday	Primary probe hybridization – 72 h
Day 3 - 4	Saturday, Sunday	(Continue primary probe hybridization)

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Day 5	Monday	Post-hybridization wash Probe securing Amplification
Day 6	Tuesday	Image on GenePS

Guidepoints

Multiple stopping points are available for storing the sample and resuming processing later. These stopping points are indicated by the STOP icon. Refer to the complete protocol below for guidance on these stopping points.

At all stopping points, it is recommended to add RNase inhibitor for extended storage periods to prevent RNA degradation.

Tips or clarifications are marked.

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Materials and Equipment

A. Spatial Genomics Materials

Refer to SDS for all handling and disposal guidelines.

Spatial Genomics Consumables	Storage
seqFISH Coverslips	4°C
Flow Cell Tops	Room Temperature

Sample Preparation Kit	Storage
Reverse Crosslinking Buffer†	-20°C
Pre-Stabilization Buffer	-20°C
Stabilization Buffer A	-20°C
Stabilization Buffer B	-20°C
Primary Wash Buffer#	-20°C
Pre-Gel Solution#	-20°C
Hydrogel Solution*	-20°C
Rinse Buffer I	-20°C
Digestion Buffer^	-20°C
Digestion Wash Buffer#	-20°C

†Only used for FFPE samples. Refer to FFPE Sample Preparation User Guide.
**Hydrogel solution needs to be supplemented with Ammonium Persulfate solution and TEMED solution to work. See details below: Section B.*
^ Digestion Buffer needs to be supplemented with Proteinase K enzyme to work. Refer to the following protocol.
 # Contains formamide – follow proper laboratory procedures.

Digestion Buffer may appear cloudy after thawing. Warm at 37°C for 15 min or until the precipitate is fully dissolved and solution is clear.

Primary Probes Kit, v175/v500/v1000¤	Storage
Pre-Hyb Mix#	-20°C
seqFISH Primary Probes#	-20°C

 # Contains formamide – follow proper laboratory procedures.
¤ Use kits of the same tier (v175, v500, or v1000) during sample preparation. So, a v175 Primary Probes Kit should be used with a v175 Amplification Kit, etc.

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Amplification Kit, v175/v500/v1000¤	Storage
Securing Components	
Primary Wash Buffer#	-20°C
Rinse Buffer II	-20°C
Pre-Securing Mix	-20°C
Securing Solution A	-20°C
Securing Solution B	-20°C
Securing Wash Buffer#	-20°C
Amplification Components	
Buffer R-1#	-20°C
Buffer R-2#	-20°C
Buffer R-3	-20°C
Buffer R-4	-20°C
Reagent R-A#	-20°C
Reagent R-B#	-20°C
Reagent R-C#	-20°C
Reagent R-D	-20°C
Reagent R-D add-in	-20°C
Reagent R-E	-20°C

 # Contains formamide – follow proper laboratory procedures.
¤ Use kits of the same tier (v175, v500, or v1000) during sample preparation. I.e., a v175 Primary Probes Kit should be used with a v175 Amplification Kit, etc.

Securing Components are used to prepare the sample before amplifying it on the SG Booster. Amplification Components are loaded into the SG Booster. Please refer to the SG Booster User Guide for details.

B. Additional Materials and Recommended Vendors

The following materials have been tested by Spatial Genomics. Equivalent options are available but may affect assay performance.

Item	Vendor	Product Number
Nuclease-Free Water	Cytiva	SH30538.02
RNase Inhibitor	VWR	PAN2615
Ammonium Persulfate (APS)*	Millipore Sigma	09913-100G
N,N,N′,N′-Tetramethylethylenediamine (TEMED)#	Millipore Sigma	T7024-25ML
Gel-Slick Solution	Lonza	50640
Proteinase K, Molecular Biology Grade	NEB	P8107S
Parafilm M	VWR	102091-164
Microscope Slide	Corning	2948-75X25
1.5 ml Microcentrifuge Tubes (RNase-free)	VWR	89000-028
PBS - Phosphate-Buffered Saline (10x) pH 7.4, RNase-free	ThermoFisher	AM9624

**Aliquot a working stock of APS to a RNase-free tube. Store both the original APS bottle and the APS aliquot on desiccant at room temperature and protect from light. Replace the original APS stock every 6 – 8 months.*
 # Make 1 mL aliquots of TEMED using RNase-free tubes and store at 4°C. Use a fresh 1 mL TEMED aliquot every week. Replace TEMED stock every 6 – 8 months.

C. Equipment

- Incubator capable of 37°C
- Ice bucket
- Oven
- Humidified chamber

A simple humidified chamber can be created by adding ~10 mL of nuclease-free water to the bottom of an empty, lidded RNase-free tip box. It is recommended to add a sheet of parafilm to lie on top of the empty tip rack, and the coverslip will lie on top of the parafilm.

D. Common Buffer Recipes

Example preparations of necessary solutions:

1. 10% APS (w/v)

Note: Make fresh every time before use.

Materials	Amount
Ammonium Persulfate	0.1 g
Nuclease free water	1 mL

Measure 0.1 g of ammonium persulfate and reconstitute it with 900 µL of nuclease-free water. Vortex thoroughly to dissolve, then verify the volume with a pipette. Add additional water to reach a final volume of 1 mL.

2. 1x PBS

Materials	Amount
10x PBS	5 mL
Nuclease free water	45 mL

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Sample Preparation

A. RNA Stabilization

1. If continuing from FFPE Tissue Collection and Preparation Guide, remove Reverse Crosslinking Buffer before proceeding.
2. Place the coverslip containing your tissue sample in a humidified chamber.

   A simple humidified chamber can be created by adding water to the bottom of an empty, lidded RNase-free tip box. Remove the tip rack and add roughly 10 mL of nuclease-free water to the bottom. Replace the tip rack and place the coverslip on top of the rack. For additional handling ease, place a clean sheet of parafilm over the tip rack and place the coverslip on top of the parafilm.

3. Add 1 mL of Pre-Stabilization Buffer to the coverslip and incubate for 1 min.
4. Warm Stabilization Buffer A and one tube of Stabilization Buffer B to room temperature.
5. Add 50 µL of Stabilization Buffer A to one tube of Stabilization Buffer B to make Mixed Stabilization Buffer. Mix gently by pipetting.
6. Remove the Pre-Stabilization Buffer by pipetting and wiping off the excess solution with a Kimwipe, avoiding the tissue area.
7. Apply 100 µL of Mixed Stabilization Buffer onto the tissue sample surface.
8. Cut a piece of parafilm slightly larger than your tissue area and use it to spread out the Mixed Stabilization Buffer to cover the whole tissue area.

   If there are bubbles around the tissue area, gently lift the parafilm and lower it with a pair of tweezers. Make sure the parafilm fits within the coverslip to avoid solution wicking away from the tissue.

9. Place the sample in the humidified chamber in a 37°C incubator for 2 h.
10. Remove the humidified chamber from the 37°C incubator and remove the parafilm.
11. Rinse the coverslip with 1 mL of 1x PBS three times.
12. Remove 1x PBS from the coverslip and pipette 1 mL of Primary Wash Buffer onto the coverslip. Incubate for 1 min, then remove Primary Wash Buffer.

B. Pre-Hydrogel Embedding

1. Pipette 50 µL of Pre-Gel Solution onto a clean parafilm larger than the coverslip.

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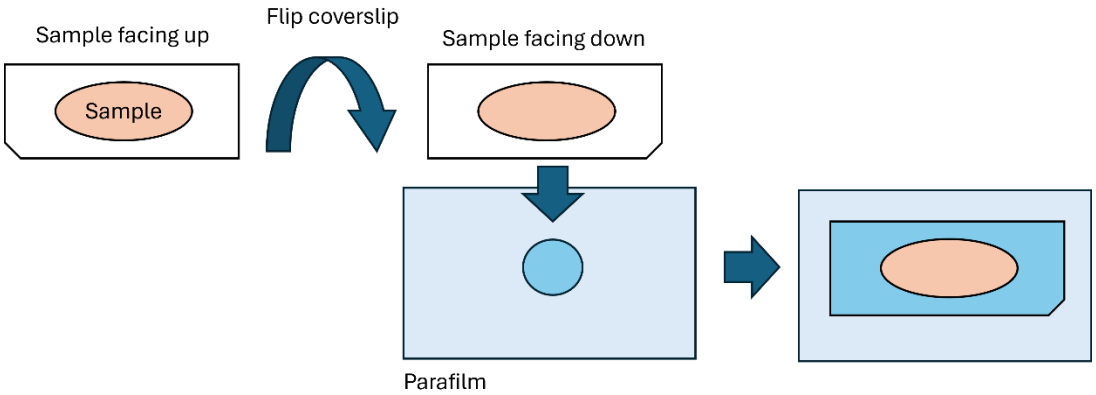
2. Invert the coverslip, with the sample facing the parafilm, and gently lay down the sample to spread out the Pre-Gel Solution.



If there are bubbles around the tissue area, gently lift the coverslip and lower it. Repeat as necessary.

3. Place the chamber containing your sample in a 37°C incubator for 30 min.

STOP

If needed, though not preferred, you may incubate the sample with the Pre-Gel Solution in a 37°C incubator overnight. Ensure that the sample is in a humidified chamber and parafilm is covering the entire tissue area.

4. Remove the chamber from the incubator and rinse the coverslip three times with 1 mL of Rinse Buffer I. On the third wash, allow the rinse buffer to incubate for 5 min.

C. Hydrogel Embedding

1. Prepare a Gel-Slick functionalized slide by pipetting 1 mL of the Gel-Slick solution onto a microscope slide. Allow this solution to sit on the slide for at least 5 min at room temperature. Pipette off excess solution and wipe excess with a Kimwipe until the slide is fully dry.

   If the Gel-Slick evaporates too quickly, it may be necessary to incubate the slide with Gel-Slick in a petri dish to prevent evaporation.

2. Thaw one tube of Hydrogel Solution and keep on ice throughout the subsequent steps. Mix well by pipetting gently.
3. Remove the Rinse Buffer I solution from the coverslip as much as possible.
4. Pipette 400 µL of the Hydrogel Solution onto a clean parafilm larger than the coverslip. Keep the remaining Hydrogel Solution on ice.

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5. Invert the coverslip, with sample facing the parafilm, and gently lay down the sample to spread out the Hydrogel Solution. Incubate for at least 5 min at RT.
6. During this time, prepare the Gelation Mix in a clean 1.5 mL tube by mixing the following components in the order listed:

Gelation Mix

Materials	Volume
Hydrogel Solution	495 µL
100% TEMED	1 µL
10% APS	4 µL

Make APS fresh, as described.

Keep the Gelation Mix on ice throughout this step.

7. Pipette 400 µL of the Gelation Mix onto a second clean parafilm larger than the size of the coverslip.
8. Lift the coverslip, with sample side facing down, and gently lay down the sample to spread out the Gelation Mix. Incubate for only 1 min.
9. Pipette 20 µL of the Gelation Mix on the Gel-Slick functionalized microscope slide in a humidified chamber. Lift the coverslip, and with the sample side facing down, gently spread out the 20 µL of Gelation Solution by sandwiching the solution with the coverslip. Avoid bubbles. Complete this step quickly before the gel begins to set.

If there are bubbles on or near the tissue area, gently lift and lower the coverslip using tweezers or with your hand to maneuver the bubbles away from the tissue. If the bubbles persist, the Gelation Mix can be wiped off the Gel-Slick slide and this step can be repeated.

10. Transfer the sample to a humidified chamber and let the sample incubate for 1.5 h at room temperature to allow gelation. Pipette the remaining 400 µL Gelation Mix off the parafilm back into the 1.5 mL tube to monitor the gelation process.
11. After 1.5 h gelation, gently remove the coverslip from the microscope slide using a razor blade. Be sure not to touch the tissue. Wipe off any excess solution on the bottom of the coverslip with a Kimwipe.
12. Place the coverslip facing up on top of the flow cell top (the back of the coverslip should be against the flow cell glass surface) so that you can see the shape of the flow cell in preparation for gel trimming.

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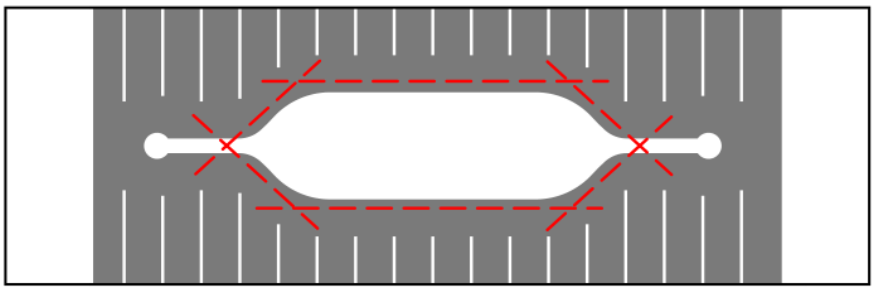
13. Cut away excess gel outside of the imaging area of the flow cell with approximately 1 mm margin. Wipe off the excess solution on the coverslip. Do this step in less than 10 min to avoid drying out the hydrogel.



14. Assemble the flow cell by adhering the flow cell top and coverslip, with the sample facing into the flow cell chamber.

> The Flow Cell Assembler can be used to aid this process.

STOP If needed, you may store the sample in Rinse Buffer I with RNase Inhibitor at 4°C for up to one week before proceeding to the next step.

15. Once the flow cell is assembled, flow in 70 µL of Digestion Buffer.

> Digestion Buffer may appear cloudy after thawing. Warm at 37°C for 15 min or until the precipitate is fully dissolved and solution is clear.

16. Clean the surface and bottom of the assembled flow cell with a Kimwipe sprayed with 70% ethanol.

> Flow cells can be labeled at this stage to differentiate between multiple samples. If using a sticker label, place the sticker away from the imaging area and ports of the flow cell.

D. Tissue Digestion

1. Make Digestion Mix by mixing the following:

Digestion Mix

Materials	Volume
Digestion Buffer	99 µL
Proteinase K, Molecular Biology Grade	1 µL

2. Remove the Digestion Buffer from the flow cell.
3. Pipette in 70 µL of the Digestion Mix. Seal the ports of the flow cell with parafilm and place the flow cell in a humidified chamber.
4. Incubate the sample for 1-24 h at 37°C, as determined best for your tissue. An overnight digestion time of 16-24 h at 37°C is typically fine and convenient. However, shorter incubation times of 1-16 h are sometimes helpful with more fragile samples or simply to decrease sample preparation time. Please see example digestion conditions in the Troubleshooting Guide at the end.

> Preparing a sample to check RNA quality with the Validation Kit is one good way to test digestion conditions. Over-digested samples typically have poor RNA quality and warped nuclei.

5. After the incubation, remove the Digestion Mix from the sample.
6. Pipette in 70 µL of the Digestion Wash Buffer and incubate at room temperature for 15 min. **Repeat** this step once.

STOP If needed, you may store the sample in Rinse Buffer I with RNase Inhibitor at 4°C for up to one week before proceeding to the next step.

E. Primary Probe Hybridization

> If using a Validation Kit, substitute Validation Primary Probes for seqFISH Primary Probes in the following steps.

1. Remove the Digestion Wash Buffer and pipette in 70 µL of Pre-Hyb Mix.
2. Place the flow cell in a humidified chamber and incubate at 37°C for 30 min.
3. Denature 70 µL of seqFISH Primary Probes at 90°C for 3 min. Allow to cool at room temperature for 1 min.
4. Completely remove Digestion Wash Buffer from the flow cell to avoid dilution of the subsequent seqFISH Primary Probes.
5. Pipette the denatured seqFISH Primary Probes into the flow cell.
6. Cover the inlets with parafilm. Place the flow cell in a humidified hybridization chamber.

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7. Place in a 37°C incubator overnight (16 h) or up to 72 h.

> Longer incubation times are recommended for larger panels:
> v175 panel – 16 h or longer; v500/v1000 panel – 36 h or longer

F. Post-Hybridization Wash

1. Remove the seqFISH Primary Probes mix from the flow cell.
2. Wash with 70 µL of Primary Wash Buffer three times.
3. On the third wash, incubate the sample in Primary Wash Buffer at 37°C for 30 min.

> The flow cell should remain in the humidified hybridization chamber during all incubation steps.

4. After 30 min, **repeat** steps 2 and 3 once to wash the sample again.
5. Remove Primary Wash Buffer.
6. Rinse with 70 µL of Rinse Buffer II three times.

STOP

If needed, you may store the sample in Rinse Buffer I with RNase Inhibitor at 4°C for up to one week before proceeding to the next step.

G. Probe Securing

1. Remove Rinse Buffer II and pipette in 70 µL of the Pre-Securing Mix.
2. Incubate at room temperature for 30 min.
3. Make Complete Securing Mix by mixing the following:

Complete Securing Mix

Materials	Volume
Securing Solution A	68.25 µL
Securing Solution B	6.75 µL

4. Remove the Pre-Securing Mix and pipette in 70 µL of the Complete Securing Mix. Incubate for 3 h at room temperature.
5. Remove the Complete Securing Mix.
6. Rinse with 70 µL of Securing Wash Buffer at room temperature 3 times.

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7. On the third wash, incubate the sample in Securing Wash Buffer at room temperature for 10 min.
8. After 10 min, **repeat** steps 6 and 7 once to wash the sample again.
9. Remove the Securing Wash Buffer and pipette in 70 µL of Rinse Buffer II. The sample is now ready to run on the SG Booster.

STOP

If needed, you may store the sample in Rinse Buffer I with RNase Inhibitor at 4°C for up to one week before proceeding to the next step.

H. Amplification

1. Follow the SG Booster User Guide for amplification instructions.
2. After amplification is complete, the sample is almost ready for imaging.
    a. If validating your sample to check its RNA quality, proceed to Sample Validation User Guide – Section D: Validation Test Labeling
    b. If setting up a seqFISH experiment, proceed to GenePS User Guide – Section 4: Setting Up a seqFISH Experiment

Troubleshooting Guide

Observation	Suggestions
Section falls off the slide	• If slide includes multiple sections, make sure sections are not overlapping • Ensure fixation buffer is fresh (made the same day) • Extend fixation time • For FFPE samples, increase baking time up to 1 hour
Gel does not separate from Gel-Slick slide	• Check that the microscope slide is not charged • Try a longer incubation time to allow the gel to fully solidify
Parts of the tissue are not embedded in the hydrogel	• Tissue that is not embedded in hydrogel will not be processed correctly; Repeat with a new section
Tissue is overly digested	• Try reducing the Digestion Mix (ProK) digestion incubation time • Reduce ProK concentration used in the Digestion Mix • Reduce digestion temperature to room temperature
Nuclei are highly warped and appear to be falling apart	• Tissue is likely over-digested; See associated suggestions
Bad RNA quality	• Ensure that you are working in a RNase-free environment • Ensure that RNase inhibitor is added to storage buffer at all stopping points • Ensure samples are collected efficiently and cleanly so as to minimize RNA degradation • Test samples with the Validation Kit to confirm good RNA quality before more in-depth experiments

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Example Digestion Conditions for Tissues

The listed tissue types and conditions are intended to illustrate example digestion conditions that have worked. These conditions are not the only ones that could work for a tissue type. For example, in many cases, shorter digestion times may work but were not tried.

Organism	Fixative	Tissue type	Digestion time	Temperature
Mouse	FF/FFPE	Liver	16–24 h	37°C
Mouse	FF/FFPE	Kidney	16–24 h	37°C
Mouse	FF/FxF	Brain	16–24 h	37°C
Mouse	FF/FxF	Spleen	16–24 h	37°C
Mouse	FF/FFPE	Embryo E12.5	16–24 h	37°C
Mouse	FFPE	Embryo E16.5	1 h	37°C
Mouse	FxF	Heart	16–24 h	37°C
Mouse	FxF	Pancreas	16–24 h	37°C
Mouse	FF/FFPE	Testis	16–24 h	37°C
Mouse	FF	Ovary	16–24 h	37°C
Human	FF	Skin	16–24 h	37°C / 47°C
Human	FF/FFPE	Breast cancer	16–24 h	37°C
Human	FF	Brain	16–24 h	37°C/ 47°C
Human	FFPE	Lung cancer	16–24 h	37°C
Human	FFPE	Colon cancer	16–24 h	37°C
Human	FxF	Embryonic kidney	16–24 h	37°C
Human	FF	Embryonic liver	16–24 h	37°C
Zebrafish	FFPE	Embryonic	1 h	37°C
Arabidopsis	FFPE	Bud	1–16 h	37°C

FF = Fresh Frozen, FxF = Fixed Frozen, FFPE = Formalin Fixed Paraffin Embedded

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